

(1.1) Budeme dokazovať na dvoch častiach

$$\bullet \operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$\begin{aligned} \|u+v\|^2 &= \langle u+v, u+v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle = \|u\|^2 + \|v\|^2 + \underbrace{\langle u, v \rangle + \overline{\langle u, v \rangle}}_{\substack{\text{prosta} \\ u+\bar{v} = 2\operatorname{Re}(u)}} \\ &= \|u\|^2 + 2\operatorname{Re}(\langle u, v \rangle) + \|v\|^2 \end{aligned}$$

$$\begin{aligned} \|u-v\|^2 &= \langle u-v, u-v \rangle = \langle u, u \rangle + \langle u, -v \rangle + \langle -v, u \rangle + \langle -v, -v \rangle = \|u\|^2 + \|v\|^2 - \langle u, v \rangle \\ &\quad - \overline{\langle u, v \rangle} = \|u\|^2 - 2\operatorname{Re}(\langle u, v \rangle) + \|v\|^2 \\ &\quad \uparrow \\ &\quad \text{prosta vyplývajú z prvej rovnice} \end{aligned}$$

$$\Rightarrow \frac{1}{4} \cdot (\|u\|^2 - \|u\|^2 + \|v\|^2 - \|v\|^2 + 2\operatorname{Re}(\langle u, v \rangle) + 2\operatorname{Re}(\langle u, v \rangle)) = \frac{1}{4} \cdot 4\operatorname{Re}(\langle u, v \rangle)$$

$$\Rightarrow \operatorname{Re}(\langle u, v \rangle) = \operatorname{Re}(\langle u, v \rangle) \quad \square$$

$$\bullet \operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$

$$\begin{aligned} \|u-iv\|^2 &= \langle u-iv, u-iv \rangle = \|u\|^2 - i\langle u, v \rangle + i\overline{\langle u, v \rangle} + \|v\|^2 \\ &= \|u\|^2 + i \cdot \underbrace{(\operatorname{Re}(\langle u, v \rangle) + i\operatorname{Im}(\langle u, v \rangle))}_{x+iy} + i(\operatorname{Re}(\langle u, v \rangle) - i\operatorname{Im}(\langle u, v \rangle)) + \|v\|^2 \\ &\quad + \|v\|^2 = \|u\|^2 - i\operatorname{Re}(\langle u, v \rangle) + \operatorname{Im}(\langle u, v \rangle) + i\operatorname{Re}(\langle u, v \rangle) + \operatorname{Im}(\langle u, v \rangle) + \|v\|^2 \\ &= \|u\|^2 + 2\operatorname{Im}(\langle u, v \rangle) + \|v\|^2 \end{aligned}$$

$$\begin{aligned} \hookrightarrow \|iv\|^2 &= \langle iv, iv \rangle \\ &= i \cdot (-i) \cdot \langle v, v \rangle \\ &= \langle v, v \rangle \end{aligned}$$

$$\begin{aligned} \|u+iv\|^2 &= \langle u+iv, u+iv \rangle = \|u\|^2 + i(\operatorname{Re}(\langle u, v \rangle) + i\operatorname{Im}(\langle u, v \rangle)) - i(\operatorname{Re}(\langle u, v \rangle) - i\operatorname{Im}(\langle u, v \rangle)) \\ &\quad + \|v\|^2 = \|u\|^2 + i\operatorname{Re}(\langle u, v \rangle) - \operatorname{Im}(\langle u, v \rangle) - i\operatorname{Re}(\langle u, v \rangle) - \operatorname{Im}(\langle u, v \rangle) + \|v\|^2 \\ &= \|u\|^2 + (-2\operatorname{Im}(\langle u, v \rangle)) + \|v\|^2 \end{aligned}$$

$$\Rightarrow \frac{1}{4} \cdot (\|u\|^2 - \|u\|^2 + \|v\|^2 - \|v\|^2 + 2\operatorname{Im}(\langle u, v \rangle) - (-2\operatorname{Im}(\langle u, v \rangle))) = \frac{1}{4} \cdot 4\operatorname{Im}(\langle u, v \rangle)$$

$$\Rightarrow \operatorname{Im}(\langle u, v \rangle) = \operatorname{Im}(\langle u, v \rangle) \quad \square$$